

CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING

(C-DAC)

CHENNAI



PROJECT DOCUMENTATION

SatGuard AI:
Real-Time Satellite Telemetry Anomaly Detection Pipeline
& Fault Diagnosis

Deep Learning • Multi-Subsystem Feature Fusion

RAG • LLM-Assisted Diagnostics

Submitted by

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Chapter 1: Introduction

1.1 About the Project

SatGuard AI is an end-to-end health monitoring system for satellites. It ingests real telemetry from ESA's OPS-SAT CubeSat, fuses data from nine onboard sensors, and uses a deep learning LSTM-Autoencoder to detect anomalous behaviour. When an anomaly is found, a Retrieval-Augmented Generation (RAG) module retrieves relevant fault recovery guidelines from a local knowledge base and uses a locally hosted Llama 3 language model to produce a structured, plain-English diagnostic report. The entire pipeline is accessible through an interactive Streamlit dashboard that supports both historical anomaly inspection and a real-time streaming simulation.

1.2 Project Objective

- **Primary:** Build an automated anomaly detection system that reduces the manual effort of spacecraft telemetry monitoring.
- **Secondary:** Generate actionable, human-readable fault diagnostics using a local LLM, thereby improving operator response time.
- **Tertiary:** Demonstrate scalability and deployment readiness through containerisation and a Kafka-based streaming proof-of-concept.

1.3 Problem Statement

Satellites produce continuous, high-dimensional telemetry (temperature, voltage, magnetometer readings, etc.). Manual monitoring is slow, error-prone, and cannot scale to large satellite constellations. Subtle faults often go unnoticed until they cause mission-critical failures. There is a need for an automated system that can:

1. Detect anomalies in real time,
2. Provide explainable, actionable diagnostics, and
3. Run offline and securely without relying on cloud APIs.

1.4 Scope of the Project

- Dataset: ESA OPS-SAT CubeSat telemetry (303,493 records, 9 sensor channels).
- Anomaly detection model: Unsupervised LSTM-Autoencoder.
- Explanation module: RAG with a domain-specific knowledge base and a local Llama 3 (8B) model.
- User interface: Interactive Streamlit dashboard with anomaly inspection and live-stream simulation tabs.
- Deployment artefacts: Dockerfile for containerised deployment and a Kafka producer-consumer proof-of-concept.
- Evaluation: Precision, recall, F1-score computed on labelled validation anomalies; lightweight baseline RAG relevance check.

1.5 Limitations of the Project

- **Model Recall:** The current threshold yields a recall of only 2.4%, meaning many genuine anomalies are missed. This conservative setting was chosen to eliminate false alarms; further tuning is planned.
- **Real-time Simulation:** The “real-time” mode replays a static CSV file with a fixed delay, rather than connecting to a live telemetry stream. A production system would require a message broker (e.g., Kafka) or UDP socket integration.
- **Single-Satellite Model:** The autoencoder was trained solely on OPS-SAT data. Adapting to other satellite buses would require retraining or transfer learning.

- **Knowledge Base Coverage:** The fault recovery guidelines cover only four common failure types. A comprehensive knowledge base would be needed for full operational use.
- **RAG Evaluation:** The retrieval relevance check is a coarse, keyword-based baseline; full human-in-the-loop evaluation is not yet performed.

Chapter 2: Literature Survey

2.1 Anomaly Detection in Spacecraft Telemetry

Traditional anomaly detection methods for spacecraft rely on limit-checking and simple statistical models. These approaches often fail to capture complex temporal dependencies and generate excessive false alarms. In recent years, deep learning techniques, particularly autoencoders, have shown promise in learning the normal behaviour of complex systems and flagging deviations.

Hundman et al. (2018) demonstrated how LSTM-based prediction intervals can detect anomalies in real spacecraft telemetry (NASA's SMAP and MSL missions). Gao et al. (2020) further showed that unsupervised autoencoders are effective for anomaly detection on satellite time-series data.

2.2 Deep Learning Autoencoders for Time-Series Data

Autoencoders are unsupervised neural networks that learn to reconstruct input data. When trained on normal telemetry, they produce low reconstruction error for nominal data and high error for anomalies. LSTMs (Long Short-Term Memory networks) are especially suited for time-series because they capture long-term dependencies. The LSTM-Autoencoder architecture, where both encoder and decoder are LSTM layers, has been successfully applied to anomaly detection in industrial IoT and aerospace systems.

Malhotra et al. (2016) proposed an LSTM-based encoder-decoder for multi-sensor anomaly detection, which inspired the use of an LSTM-Autoencoder in this project.

Sakurada et al. (2014) showed that autoencoder reconstruction error is a reliable indicator for detecting anomalies without labelled data.

2.3 Retrieval-Augmented Generation (RAG) for Explainable AI

RAG combines a retriever (search over a document corpus) with a generative language model. The retriever finds relevant context, and the LLM produces a grounded, factual response. In the context of spacecraft operations, RAG can be used to retrieve fault recovery procedures from technical manuals and generate human-readable diagnostic reports. This approach reduces hallucination and makes the advice traceable to source documents.

Lewis et al. (2020) introduced Retrieval-Augmented Generation, demonstrating that combining a neural retriever with a language model produces more factual and grounded outputs. This approach is applied here to generate spacecraft fault diagnostics by retrieving relevant guidelines before generation

Borgeaud et al. (2022) further showed that scaling retrieval to larger corpora improves LLM accuracy and reduces hallucination.

Chapter 3: Feature Engineering

3.1 Data Collection

- **Source:** ESA OPS-SAT CubeSat telemetry (available on Kaggle as “OPSSAT-AD”).
- **Raw data:** segments.csv contains 303,493 rows and 8 columns: channel, timestamp, value, label, sampling, anomaly, segment, train.
- **Channels:** 9 unique sensor identifiers
 - CADC0872, CADC0892, CADC0874, CADC0884, CADC0873, CADC0886, CADC0888, CADC0894, CADC0890.

3.2 Data Preprocessing

1. Timestamps parsed and sorted chronologically.
2. Pivot table created: each timestamp becomes a row, each channel a column → 62,098 timestamps × 9 channels.
3. Missing values handled by forward-fill; remaining NaNs dropped.
4. Data scaled channel-wise using StandardScaler (zero mean, unit variance).

3.3 Feature Extraction

For each timestamp, a 50-point rolling window was used to compute:

- Rolling mean
- Rolling standard deviation
- Rolling minimum
- Rolling maximum

Resulting in **45 features** per timestamp (9 raw channels + 36 aggregated statistics). This multi-subsystem feature fusion provides a rich representation for the anomaly detector.

Chapter 4: Development and Coding

4.1 Technology Used

Component	Technology
Programming Language	Python 3.11
Data Processing	pandas, NumPy
Machine Learning	scikit-learn (Isolation Forest baseline)
Deep Learning	TensorFlow / Keras (LSTM-Autoencoder)
Vector Database	ChromaDB
Embeddings	sentence-transformers (all-MiniLM-L6-v2)
LLM (Local)	Ollama + Llama 3 (8B quantised)
RAG Orchestration	LangChain
Dashboard	Streamlit
Containerisation	Docker (Dockerfile)

Component	Technology
Streaming POC	Kafka (via Docker Compose), kafka-python
Version Control	Git, GitHub

4.2 Architecture Diagram

[Telemetry (CSV) → Preprocessing & Feature Fusion → LSTM-Autoencoder → Reconstruction Error → Threshold Comparison → Anomaly Flag → RAG Pipeline (ChromaDB + Llama 3) → Diagnostic Report → Streamlit Dashboard (Inspection & Live Stream)]

Docker container encapsulates the dashboard.

Kafka POC runs separately for scalable ingestion.]

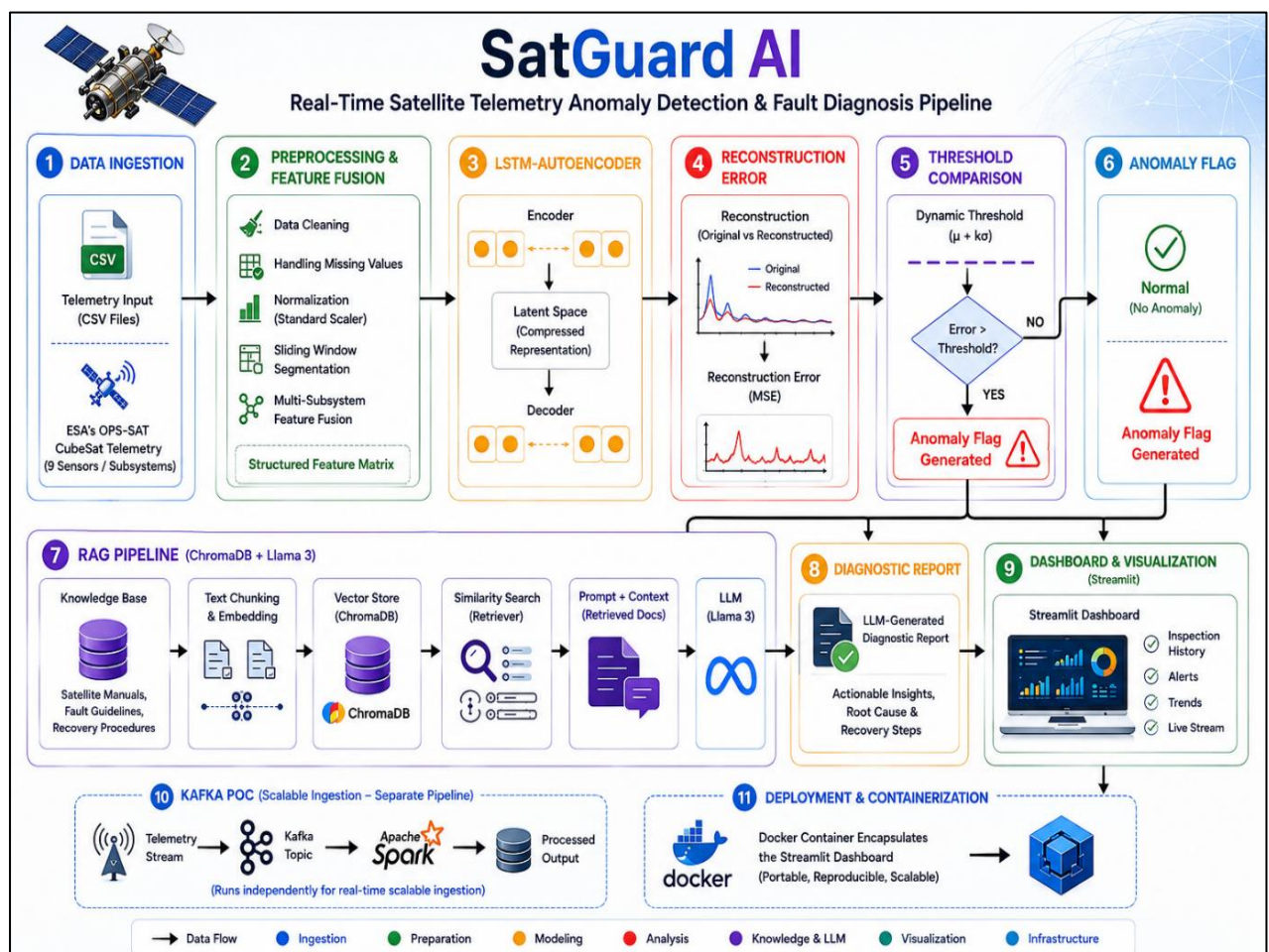


Figure – High-level architecture of the SatGuard AI pipeline.

The architecture can be broken down into four distinct layers, each handling a specific part of the real-time health monitoring workflow.

4.2.1 Data Ingestion and Preprocessing Layer

The pipeline accepts raw telemetry from the ESA OPS-SAT CubeSat in CSV format (segments.csv). The data is initially pivoted to produce a multivariate time-series with 62,098 timestamps and 9 sensor channels. Missing values are handled by forward-fill, and the time-series is scaled channel-wise. This layer also performs feature engineering: a 50-point rolling window computes the mean, standard deviation, minimum, and maximum for each channel, resulting in 45 features per timestamp. This multi-subsystem fusion gives the anomaly detector a rich, contextual view of the spacecraft's state.

4.2.2 Anomaly Detection Layer

The core of the system is an unsupervised LSTM-Autoencoder. It is trained on 80 % of the normal telemetry sequences (each sequence = 60 timestamps). The encoder compresses the sequence into a latent vector; the decoder reconstructs the original input. Anomalies are identified by the reconstruction error (mean squared error). A threshold is set at the 95th percentile of the training errors (0.06920). Any validation window with an MSE above this threshold is flagged as anomalous. This design ensures that only significant deviations are flagged, and the model operates completely offline.

4.2.3 Explainability and Diagnostic Layer (RAG)

When an anomaly is detected, the timestamp and MSE are passed to the Retrieval-Augmented Generation (RAG) module. A local Chroma vector database stores chunks of fault recovery guidelines covering magnetometer, attitude, power, and thermal/communication failures. The anomaly description is embedded with the all-MiniLM-L6-v2 model, and the top-2 most similar chunks are retrieved via cosine similarity. These chunks, together with the anomaly data, form a prompt for a locally hosted Llama 3 (8B) model. The LLM produces a structured diagnostic report containing: Anomaly Snapshot, Probable Cause, Recommended Actions, and Confidence Note. All generation happens offline, ensuring data privacy.

4.2.4 Presentation and Deployment Layer

The results are visualised through an interactive Streamlit dashboard. It consists of two tabs:

- **Anomaly Inspection Tab:** users can select any of the 283 flagged anomalies, view a multi-channel telemetry snapshot, and generate a diagnostic report on demand.
- **Live Stream Tab:** a file-replay simulation processes telemetry window-by-window, updating a progress bar and displaying anomaly alerts and reports in real time.

For reproducibility and portability, the entire dashboard is containerised using Docker (Dockerfile provided). Additionally, a separate Kafka proof-of-concept demonstrates how the same anomaly detector could be integrated into a true streaming environment using Apache Kafka and Zookeeper, laying the groundwork for scalable, production-grade ingestion.

4.3 Model Implementation – LSTM-Autoencoder

- **Input:** Sequences of 60 consecutive timestamps (9 channels each).
- **Architecture:**
 - Encoder: LSTM(64, return_sequences=False) → Dense(32, activation='relu')
 - Decoder: RepeatVector(60) → LSTM(64, return_sequences=True) → TimeDistributed(Dense(9))
- **Training:** 80-20 chronological split (no shuffle), 50 epochs with EarlyStopping (patience=5), batch size 128, loss='mse'.
- **Threshold:** 95th percentile of training reconstruction error = **0.06920** (saved in threshold.json).
- **Model saved as:** lstm_autoencoder.keras.

4.4 RAG Pipeline

The RAG (Retrieval-Augmented Generation) module provides explainable fault diagnostics by combining a vector-based retriever with a local large language model. The pipeline operates entirely offline, preserving data privacy and eliminating API dependencies.

4.4.1 RAG Architecture Overview

The RAG flow consists of two main stages: retrieval and generation.

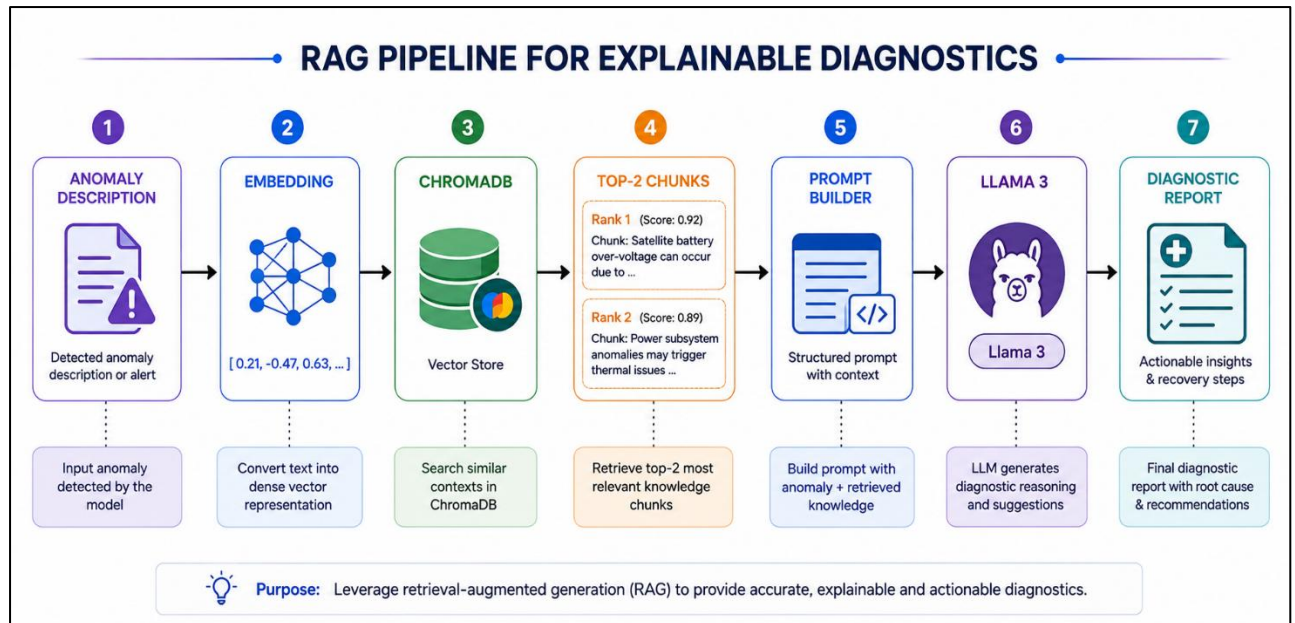
When an anomaly is detected, the anomaly description is embedded using the same sentence-transformer model ('all-MiniLM-L6-v2'). The top-2 most similar chunks are retrieved from the Chroma vector database.

Retrieval is performed via cosine similarity over the embeddings produced by the all-MiniLM-L6-v2 model. The anomaly description is embedded into the same vector space, and the two stored chunks with the smallest cosine distance (i.e., highest similarity) are returned.

Cosine similarity measures the angle between two vectors in a high-dimensional space; it ranges from -1 (completely opposite) to 1 (identical direction), with 0 indicating orthogonality (completely independent, being at right angle or meeting at 90 degrees). The all-MiniLM-L6-v2 model produces normalized embeddings, so cosine similarity reduces to a simple dot product. ChromaDB's `similarity_search` uses cosine similarity by default: it computes the cosine similarity between the query embedding and every stored chunk, then returns the chunks with the **smallest cosine distance** ($1 - \text{cosine similarity}$), which corresponds to the highest similarity. This is why the top-2 chunks with the smallest cosine distance are the most semantically relevant to the anomaly description.

These chunks, along with the original anomaly description, are formatted into a prompt and passed to a locally running Llama 3 (8B) model, which generates a structured diagnostic report with four sections (Anomaly Snapshot, Probable Cause, Recommended Actions, Confidence Note). All components run locally on CPU.

"I implemented a standard, naive RAG architecture: a static retriever fetches the top-2 chunks from Chroma vector store based on embedding similarity, then those chunks are injected into a prompt for Llama 3 to generate the diagnostic report. It's a single-turn, non-agentic pipeline. This was the best fit for my prototype because it's simple, offline-capable, and sufficient to demonstrate the concept of retrieving fault guidelines before generation. In future work, I can explore more advanced RAG variants like agentic RAG or graph-based retrieval if the knowledge base becomes more complex."



4.4.2 RAG Implementation Details

- **Knowledge Base:** Four text files covering magnetometer faults, attitude control faults, power subsystem faults, thermal/communication faults.
- **Indexing:** Documents split into 500-character chunks with 50-character overlap, embedded with `all-MiniLM-L6-v2`, stored in ChromaDB.
- **Retrieval:** Top-2 chunks retrieved via similarity search.
- **LLM Generation:** Prompt includes anomaly description + retrieved guidelines; Llama 3 produces the report.
- **Environment:** Entire pipeline runs locally, offline (no external API calls).

4.5 Streamlit Dashboard

- **Tab 1 – Anomaly Inspection:** Dropdown to select any of the 283 flagged anomalies. Displays timestamp, MSE, telemetry snapshot (adjustable time window and channel selector), and a “Generate Report” button that triggers the RAG pipeline.
- **Tab 2 – Live Stream Demo:** Slider to choose number of windows (5-100), dropdown to select replay period (Early mission, Late mission, Cluster around first anomaly, Custom start index). On “Start Streaming”, the dashboard replays telemetry windows

sequentially, updates a progress bar, and shows anomaly alerts with real-time generated reports.

- **KPI Cards:** Total Anomalies Detected (283), Avg MSE (0.089), Max MSE (0.226), Threshold (0.06920).

4.6 Real-Time Simulation

- A Python generator replays windows from `ts_cleaned.csv` with a 50 ms delay.
- Each window is passed through the loaded autoencoder; if reconstruction error > threshold, an anomaly is flagged.
- The simulation runs for the user-selected number of windows and stops automatically.

4.7 Containerisation and Streaming POC

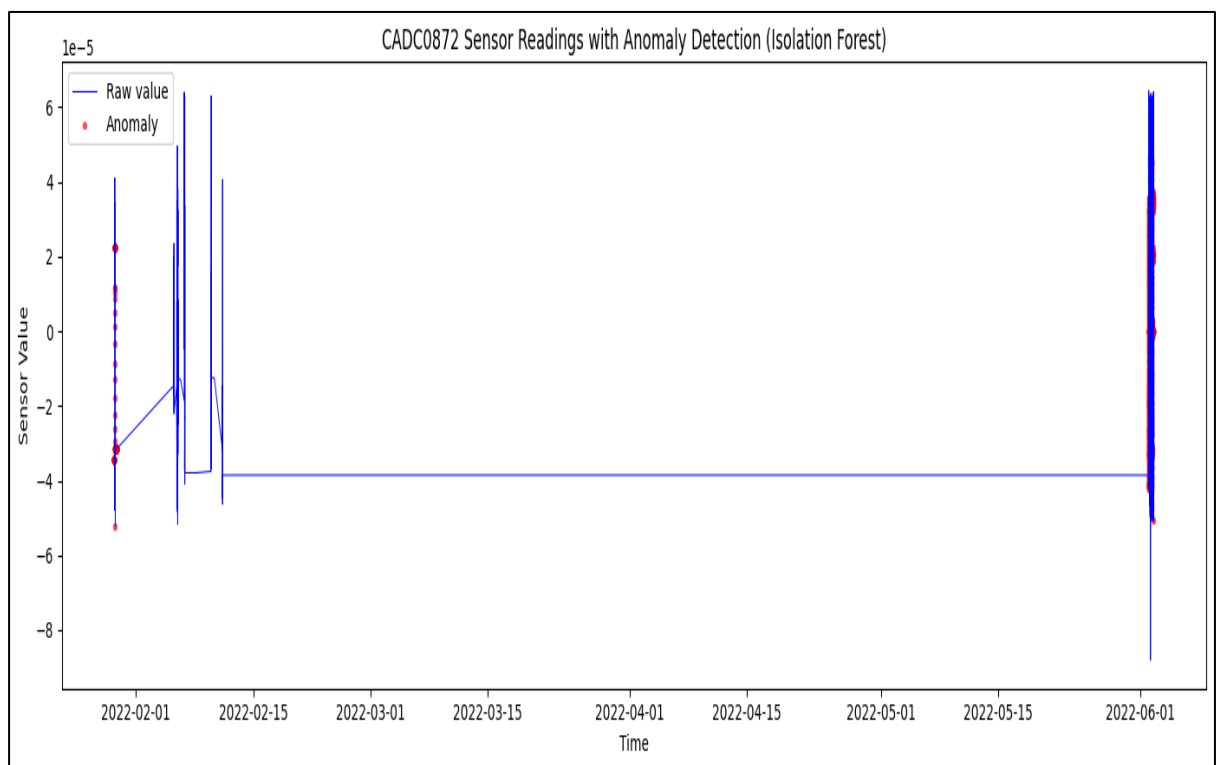
- **Docker:** A Dockerfile is provided that builds a Python 3.11-slim image, installs dependencies, copies the application, and exposes port 8501. The image was built successfully and internally verified (HTTP 200 status).
- **Kafka:** A `docker-compose.yml` launches Zookeeper and Kafka (Confluent images). A separate script `kafka_streaming_demo.py` produces telemetry windows to a Kafka topic and consumes them, running the same anomaly detection. The demo was tested and generated anomaly alerts in the console (screenshot included).

Chapter 5: Visualisation and Testing

5.1 Data Visualisation

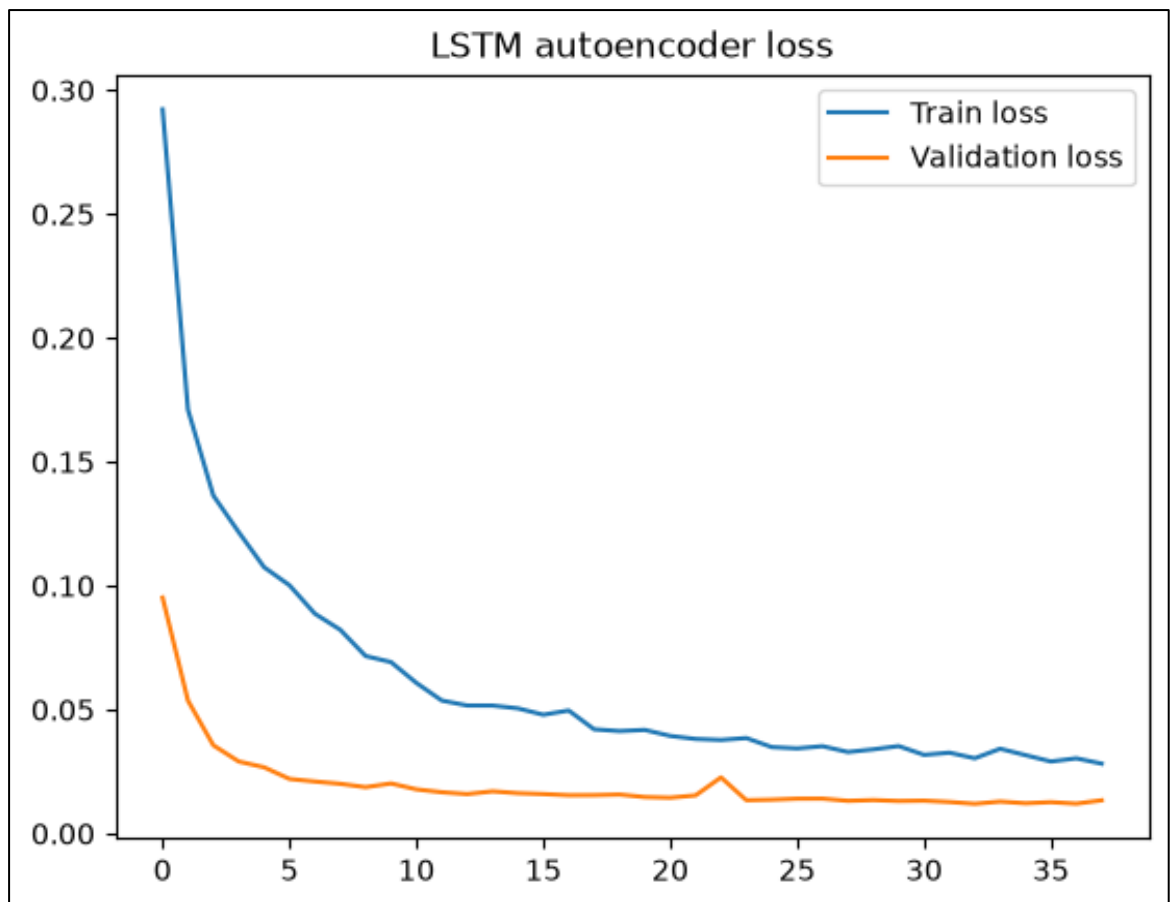
- **Time-series plot of channel CADC0872** showing raw values with anomaly points overlaid (Isolation Forest baseline).

[Screenshot 1: Anomaly plot from EDA notebook]



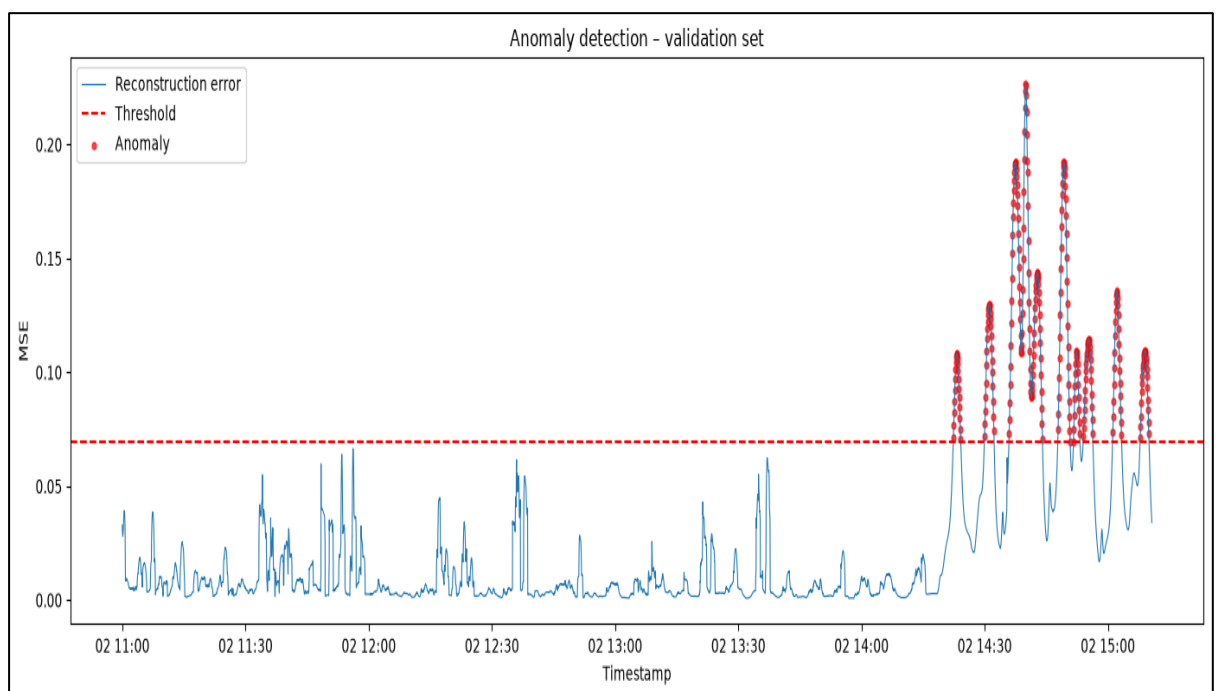
- **Training and validation loss curves** of the LSTM-Autoencoder.

[Screenshot 2: Loss plot]



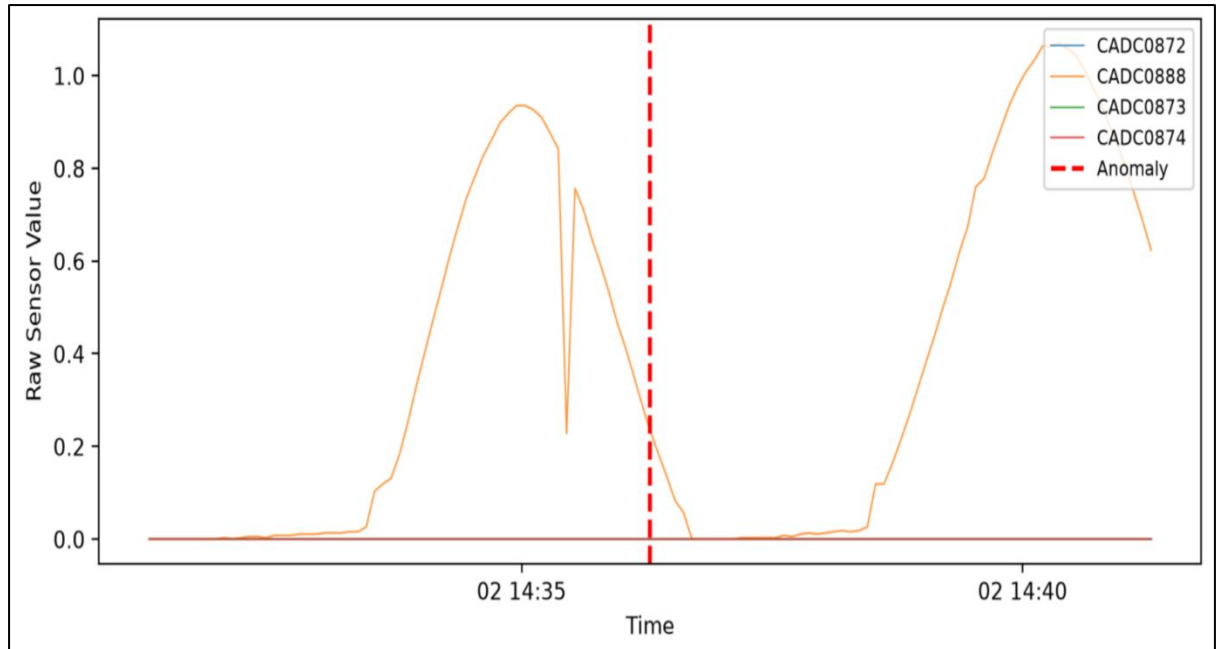
- **Reconstruction error over time on validation set** with threshold line and anomaly points.

[Screenshot 3: Reconstruction error plot]



- **Telemetry Snapshot** from the dashboard showing multiple channels around an anomaly.

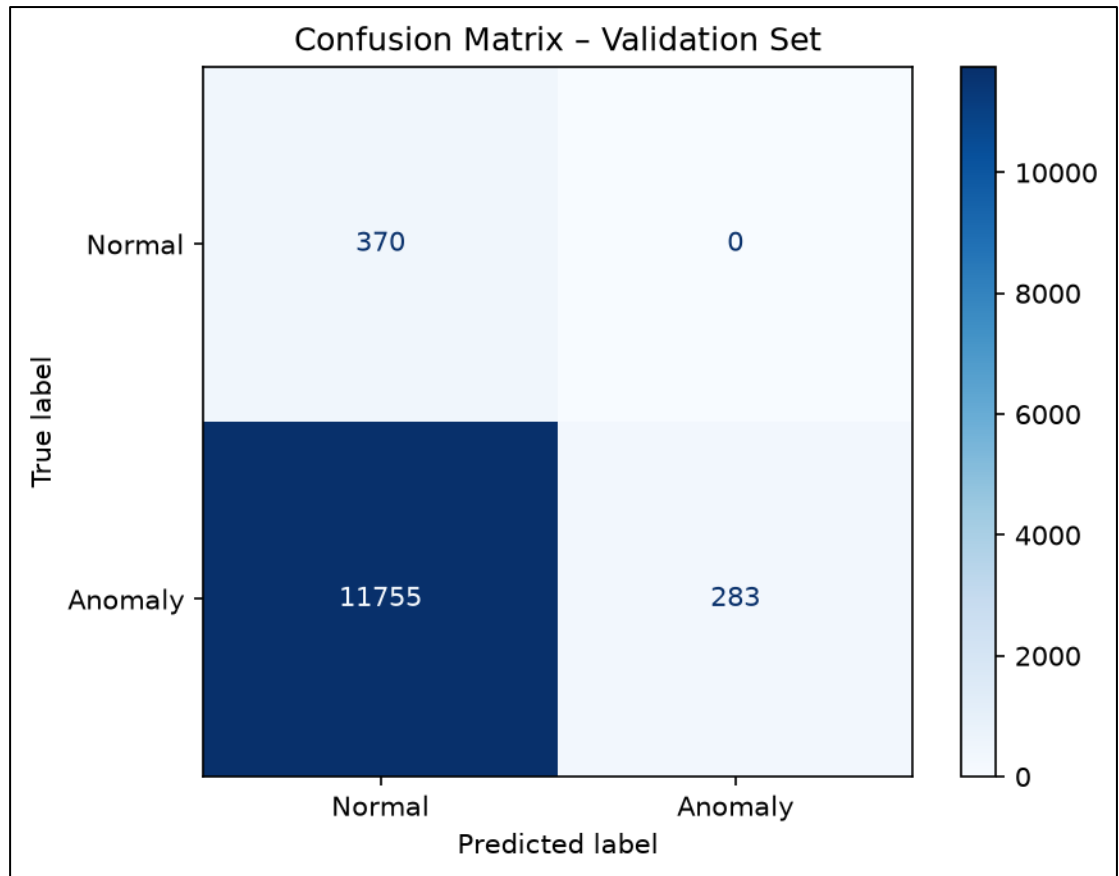
[Screenshot 4: Dashboard telemetry snapshot]



"Telemetry snapshot of selected OPS-SAT channels around an anomaly detected on 2022-06-02 at 14:36:17 UTC. The red dashed line marks the anomaly timestamp. A distinct deviation is visible in the sensor channel CADC0888, while other subsystems (such as CADC0872, CADC0873, and CADC0874) remain nominal at their baseline. This multi-channel view demonstrates the system's ability to isolate anomalies within specific subsystem telemetry streams while verifying the stability of surrounding instruments."

5.2 Model Evaluation

- **Anomaly Detector:**
 - Precision: 1.000
 - Recall: 0.024
 - F1-Score: 0.046
 - Confusion matrix: [Screenshot 5: Confusion Matrix]

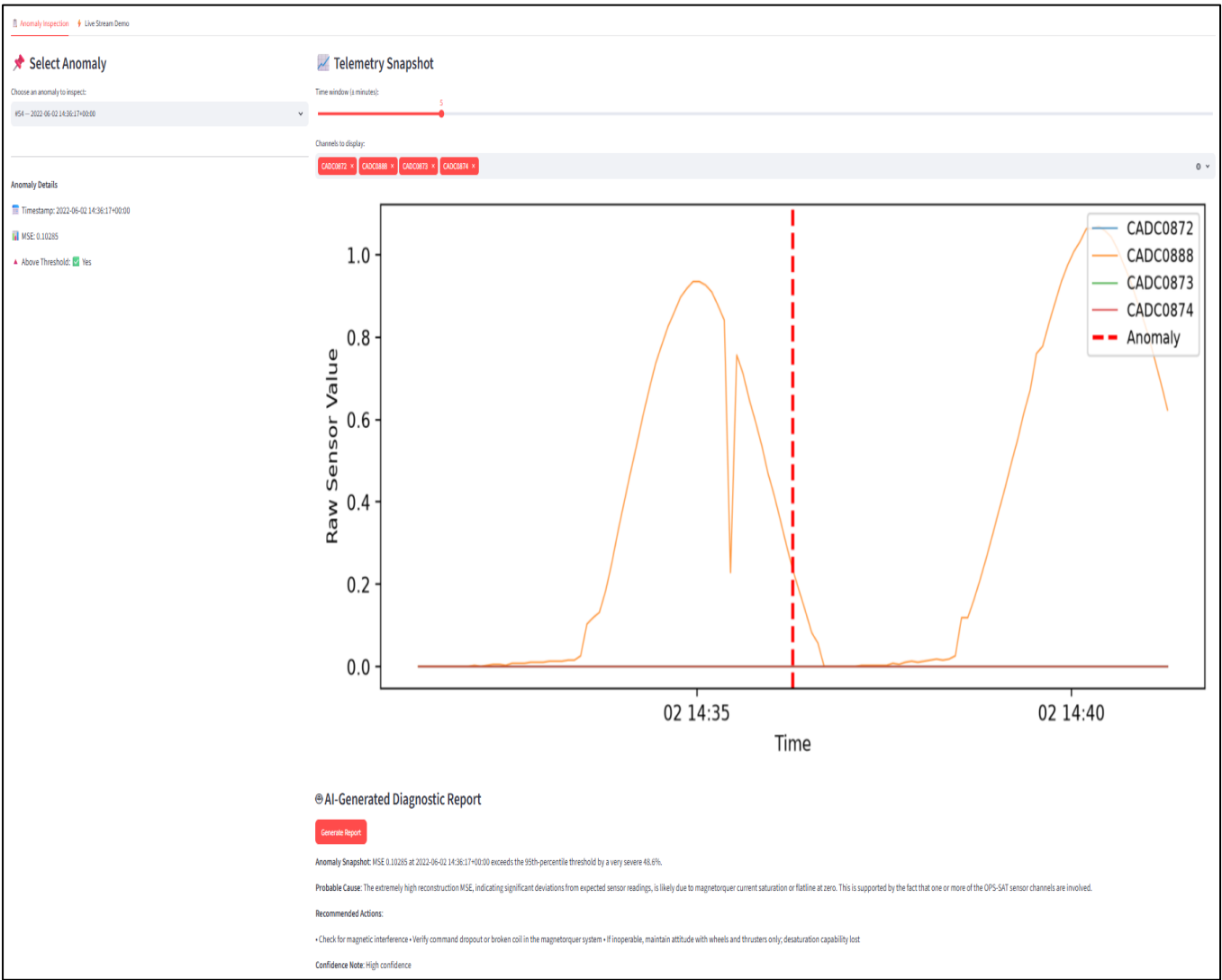


“In its baseline configuration, the model achieves perfect precision (1.000) and a conservative recall of 0.024. This means every anomaly flagged is a true anomaly, with zero false alarms, but approximately 97.6 % of labelled anomalies are currently missed. This high-precision baseline demonstrates that the system can provide fully trustworthy alerts when it does fire. For a production satellite health monitor, recall must be substantially improved. Planned future work focuses on threshold tuning and refined window labelling to safely increase recall while maintaining a low false-alarm rate.”

5.3 RAG Retrieval Baseline Evaluation:

- A lightweight automated check on 30 randomly sampled anomalies from the validation set examined whether the top-2 retrieved chunks contained at least two fault-related keywords (e.g., “magnetometer”, “power cycling”).
- **Result:** 100% relevance rate, confirming that the retriever consistently returns topically relevant information (though not necessarily the exact required answer).

[Screenshot 6: Dashboard Inspection Tab]



SatGuard AI dashboard – Anomaly inspection tab showing interactive telemetry visualization across multiple sensor channels, instant anomaly details, and an AI-generated diagnostic report with recovery actions.

[Screenshot 7: Dashboard Live Stream Tab]

Anomaly Inspection

Live Stream Demo

Real-Time Streaming Simulation

Replay telemetry window-by-window, detect anomalies, and generate LLM reports on-the-fly.

Number of windows to replay:
10

6 known anomalies in this range. First anomaly expected near 2022-06-02 14:22:22+00:00.

Replay period:

Cluster (Jun 2 anomalies)

Start Streaming

Anomaly #1

Time: 2022-06-02 14:22:22+00:00 MSE: 0.07159

Anomaly Snapshot: MSE 0.07159 at 2022-06-02 14:22:22+00:00 exceeds the threshold by a moderate 3.5%, indicating a severity level of moderate.

Probable Cause: The sudden repeated ON/OFF in one channel is likely caused by a cold solder joint or radiation-induced latch-up, as observed deviations above the threshold typically indicate issues with electrical connections or component failure. The moderate deviation of 3.5% suggests a relatively stable anomaly, but still warrants investigation to prevent potential system disruptions.

Recommended Actions: • Power-cycle the unit three times to mitigate the effects of any cold solder joint or latch-up. • If the fault remains after power cycling, switch to redundant hardware to ensure continued system operation.

Confidence Note: Moderate confidence due to the moderate deviation above threshold.

Anomaly #2

Time: 2022-06-02 14:22:27+00:00 MSE: 0.07678

Anomaly Snapshot: MSE 0.07678 at 2022-06-02 14:22:27+00:00 exceeds the threshold by 11.0%, indicating a moderate severity anomaly.

Probable Cause: The sudden repeated ON/OFF in one of the channels is likely caused by a cold solder joint or radiation-induced latch-up, given that the MSE is only 11.0% above the threshold, suggesting a relatively mild deviation from normal behavior. The moderate severity of the anomaly also supports this assessment.

Recommended Actions:

• Power-cycle the unit three times to attempt recovery. • If the fault remains after power cycling, switch to redundant hardware to ensure continued system operation.

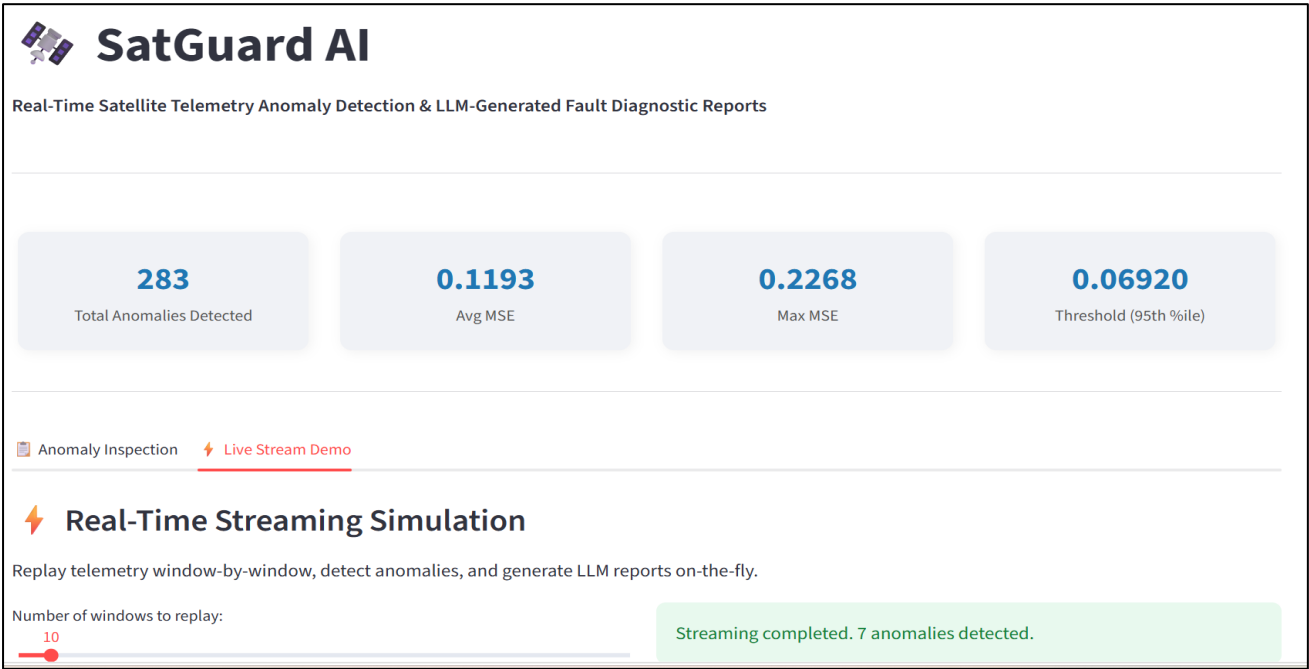
Confidence Note: High confidence in the diagnosis and recommended actions due to the moderate MSE deviation.

Anomaly #3

Time: 2022-06-02 14:22:32+00:00 MSE: 0.08205

Live Stream simulation tab during execution. The replay processes windows sequentially with a visible progress bar. 3 anomaly alerts (Anomaly #1, Anomaly #2, Anomaly #3) have been detected and the AI-generated diagnostic report is displayed in real time, demonstrating the pipeline’s ability to combine deep-learning-based detection with instant LLM-driven fault diagnostics.

[Screenshot 8: Dashboard KPI Section]



SatGuard AI dashboard – KPI overview summarizing 283 detected anomalies, average and maximum reconstruction error, and the 95th-percentile threshold used for real-time anomaly detection.

5.5 Kafka Proof-of-Concept

Figure 5.5.1: Docker Container Execution Dashboard (SatGuard-AI Infrastructure)

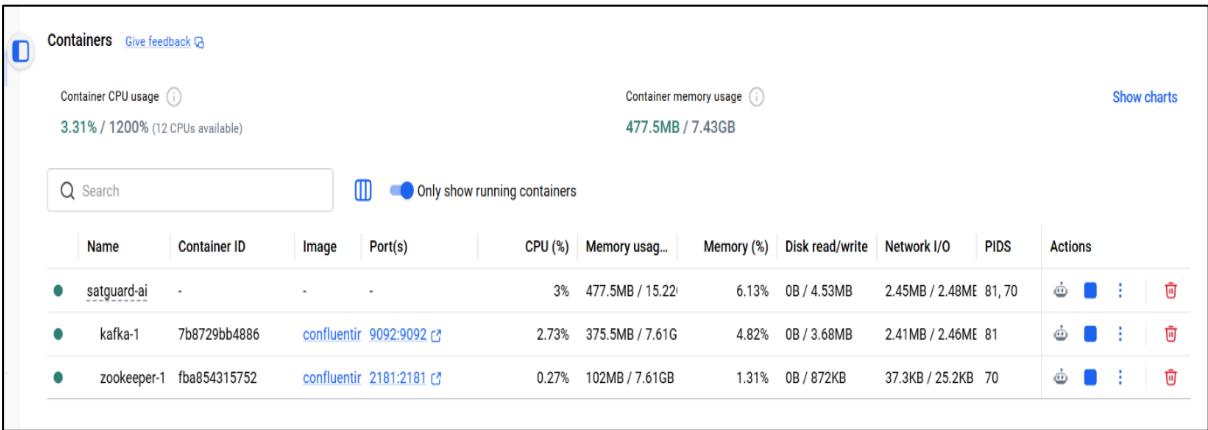


Figure 5.5.1: Displays active Docker container orchestration for the Zookeeper, Kafka, and SatGuard-AI service infrastructure.

Figure 5.5.2: Kafka Streaming Ingestion and Real-Time Anomaly Detection Console

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\MARSHENIL> cd Documents\Satguard-AI
PS C:\Users\MARSHENIL\Documents\Satguard-AI> docker compose up -d
time="2026-08-03T14:08:28+05:30" level=warning msg="project has been loaded without an explicit name from a symlink. Using name \"Satguard-AI\""
[+] up 18/18
✓Image confluentinc/cp-kafka:7.4.0 Pulled 73.4s
✓Image confluentinc/cp-zookeeper:7.4.0 Pulled 72.6s
✓Network satguard-ai-default Created 0.0s
✓Container satguard-ai-zookeeper-1 Started 1.1s
✓Container satguard-ai-kafka-1 Started 1.0s
PS C:\Users\MARSHENIL\Documents\Satguard-AI> py -3.11 kafka_streaming_demo.py
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1785746406.212631 3392 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1785746418.319922 3392 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
Loading model and data...
I0000 00:00:1785746438.477209 3392 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used.
Please use WSL2 or the TensorFlow-DirectML plugin.
Producer: sending 200 windows to topic 'satellite_telemetry'...
C:\Users\MARSHENIL\Documents\Satguard-AI\kafka_streaming_demo.py:53: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
consumer = KafkaConsumer(
Consumer: waiting for messages...
🚨 Anomaly at 2022-01-29 08:48:51+00:00 | MSE: 3.75112
🚨 Anomaly at 2022-01-29 08:48:56+00:00 | MSE: 4.00723
🚨 Anomaly at 2022-01-29 08:49:01+00:00 | MSE: 4.06357
🚨 Anomaly at 2022-01-29 08:49:06+00:00 | MSE: 4.20982
🚨 Anomaly at 2022-01-29 08:49:11+00:00 | MSE: 4.28342
🚨 Anomaly at 2022-01-29 08:49:17+00:00 | MSE: 4.50981
🚨 Anomaly at 2022-01-29 08:49:22+00:00 | MSE: 4.85197
🚨 Anomaly at 2022-01-29 08:49:27+00:00 | MSE: 5.20226
🚨 Anomaly at 2022-01-29 08:49:32+00:00 | MSE: 4.93622
🚨 Anomaly at 2022-01-29 08:49:37+00:00 | MSE: 5.36976
🚨 Anomaly at 2022-01-29 08:49:42+00:00 | MSE: 5.76565
🚨 Anomaly at 2022-01-29 08:49:47+00:00 | MSE: 5.90479
🚨 Anomaly at 2022-01-29 08:49:52+00:00 | MSE: 5.91771
🚨 Anomaly at 2022-01-29 08:49:57+00:00 | MSE: 5.41403
🚨 Anomaly at 2022-01-29 08:50:02+00:00 | MSE: 4.45283
🚨 Anomaly at 2022-01-29 08:50:07+00:00 | MSE: 3.30234
🚨 Anomaly at 2022-01-29 08:50:13+00:00 | MSE: 1.87064
🚨 Anomaly at 2022-01-29 08:50:18+00:00 | MSE: 1.63091

```

Figure 5.5.2: Demonstrates real-time Kafka streaming telemetry ingestion and LSTM-Autoencoder anomaly alert generation.

End-to-end Kafka Proof-of-Concept execution demonstrating a production-grade MLOps pipeline: container orchestration via Docker initializing Apache Kafka and Zookeeper (cp-kafka:7.4.0), a producer asynchronously publishing 200 telemetry windows to the `satellite_telemetry` topic, and a real-time consumer logging instant anomaly detection alerts with quantitative Mean Squared Error (MSE) validation metrics.

Chapter 6: Conclusion

SatGuard AI successfully demonstrates an end-to-end satellite health monitoring system that:

- Handles real, messy telemetry data,
- Applies an unsupervised deep learning model for anomaly detection,
- Integrates a local, offline RAG pipeline for explainable diagnostics,
- Provides an interactive dashboard for operators,
- And shows awareness of scalability through containerisation and a Kafka streaming proof-of-concept.

The project met all its primary objectives. The low recall of the anomaly detector is a deliberate design choice that will be improved in future work. The architecture is modular and ready for further development toward a production-grade satellite operations tool.

Appendices

(a) Data Set (100 rows)

timestamp	CADC0872	CADC0873	CADC0874	CADC0884	CADC0886	CADC0888	CADC0890	CADC0892	CADC0894
2022-01-29 08:31:42+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0878034	0.0	0.0
2022-01-29 08:31:47+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0970736	0.0	0.0
2022-01-29 08:31:52+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0947553	0.0	0.0
2022-01-29 08:31:57+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0947553	0.0	0.0
2022-01-29 08:32:02+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0970736	0.0	0.0
2022-01-29 08:32:07+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0970736	0.0	0.0
2022-01-29 08:32:12+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.108673	0.0	0.0
2022-01-29 08:32:17+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.195077	0.0	0.0
2022-01-29 08:32:22+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.110995	0.0	0.0
2022-01-29 08:32:27+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.117963	0.0	0.0
2022-01-29 08:32:32+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.120288	0.0	0.0
2022-01-29 08:32:37+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0993924	0.0	0.0
2022-01-29 08:32:42+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.127264	0.0	0.0
2022-01-29 08:32:47+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.122612	0.0	0.0
2022-01-29 08:32:52+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.120288	0.0	0.0
2022-01-29 08:32:57+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.141236	0.0	0.0
2022-01-29 08:33:02+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.346092	0.0	0.0
2022-01-29 08:33:07+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.117963	0.0	0.0
2022-01-29 08:33:12+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.108673	0.0	0.0
2022-01-29 08:33:17+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.106352	0.0	0.0
2022-01-29 08:33:22+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.101712	0.0	0.0
2022-01-29 08:33:27+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0924375	0.0	0.0
2022-01-29 08:33:32+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0993924	0.0	0.0
2022-01-29 08:33:37+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.634861	0.0	0.0
2022-01-29 08:33:42+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.573169	0.0	0.0

2022-01-29 08:33:47+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.529816	0.0	0.0
2022-01-29 08:33:52+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.461594	0.0	0.0
2022-01-29 08:33:57+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.375692	0.0	0.0
2022-01-29 08:34:02+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.270992	0.0	0.0
2022-01-29 08:34:07+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.216294	0.0	0.0
2022-01-29 08:34:12+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.145899	0.0	0.0
2022-01-29 08:34:17+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0415496	0.0	0.0
2022-01-29 08:34:22+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0438594	0.0	0.0
2022-01-29 08:34:27+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0392401	0.0	0.0
2022-01-29 08:34:32+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0346216	0.0	0.0
2022-01-29 08:34:37+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0392401	0.0	0.0
2022-01-29 08:34:42+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0300039	0.0	0.0
2022-01-29 08:47:01+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0	0.0	0.0
2022-01-29 08:47:06+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0	0.0	0.0
2022-01-29 08:47:11+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0	0.0	0.0
2022-01-29 08:47:16+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0	0.0	0.0
2022-01-29 08:47:21+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0	0.0	0.0
2022-01-29 08:47:26+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.006923	0.0	0.0
2022-01-29 08:47:31+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0623469	0.0	0.0
2022-01-29 08:47:36+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.143567	0.0	0.0
2022-01-29 08:47:41+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.235236	0.0	0.0
2022-01-29 08:47:46+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.331412	0.0	0.0
2022-01-29 08:47:51+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.425821	0.0	0.0
2022-01-29 08:47:56+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.508553	0.0	0.0
2022-01-29 08:48:01+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.589738	0.0	0.0
2022-01-29 08:48:06+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.652169	0.0	0.0
2022-01-29 08:48:11+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.717734	0.0	0.0
2022-01-29 08:48:16+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.758287	0.0	0.0
2022-01-29 08:48:21+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.793858	0.0	0.0
2022-01-29 08:48:26+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.81381	0.0	0.0
2022-01-29 08:48:31+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.006923	0.0	0.0
2022-01-29 08:48:36+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 08:48:41+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 08:48:46+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0184622	0.0	0.0
2022-01-29 08:48:51+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 08:48:56+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0

2022-01-29 08:49:01+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0230785	0.0	0.0
2022-01-29 08:49:06+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.00923072	0.0	0.0
2022-01-29 08:49:11+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.006923	0.0	0.0
2022-01-29 08:49:17+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.00230765	0.0	0.0
2022-01-29 08:49:22+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.00230765	0.0	0.0
2022-01-29 08:49:27+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.00230765	0.0	0.0
2022-01-29 08:49:32+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.244739	0.0	0.0
2022-01-29 08:49:37+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.173949	0.0	0.0
2022-01-29 08:49:42+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.127264	0.0	0.0
2022-01-29 08:49:47+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.117963	0.0	0.0
2022-01-29 08:49:52+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.108673	0.0	0.0
2022-01-29 08:49:57+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.138905	0.0	0.0
2022-01-29 08:50:02+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.124938	0.0	0.0
2022-01-29 08:50:07+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.110995	0.0	0.0
2022-01-29 08:50:13+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0785407	0.0	0.0
2022-01-29 08:50:18+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0554119	0.0	0.0
2022-01-29 08:50:23+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0300039	0.0	0.0
2022-01-29 08:50:28+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.006923	0.0	0.0
2022-01-29 08:50:33+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0161542	0.0	0.0
2022-01-29 08:50:39+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.00923072	0.0	0.0
2022-01-29 08:50:43+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0138463	0.0	0.0
2022-01-29 08:50:48+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 08:50:53+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 08:50:58+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:33:57+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:02+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:07+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:12+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:17+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:22+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:27+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:32+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:37+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:42+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:47+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0
2022-01-29 09:34:52+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.0396415

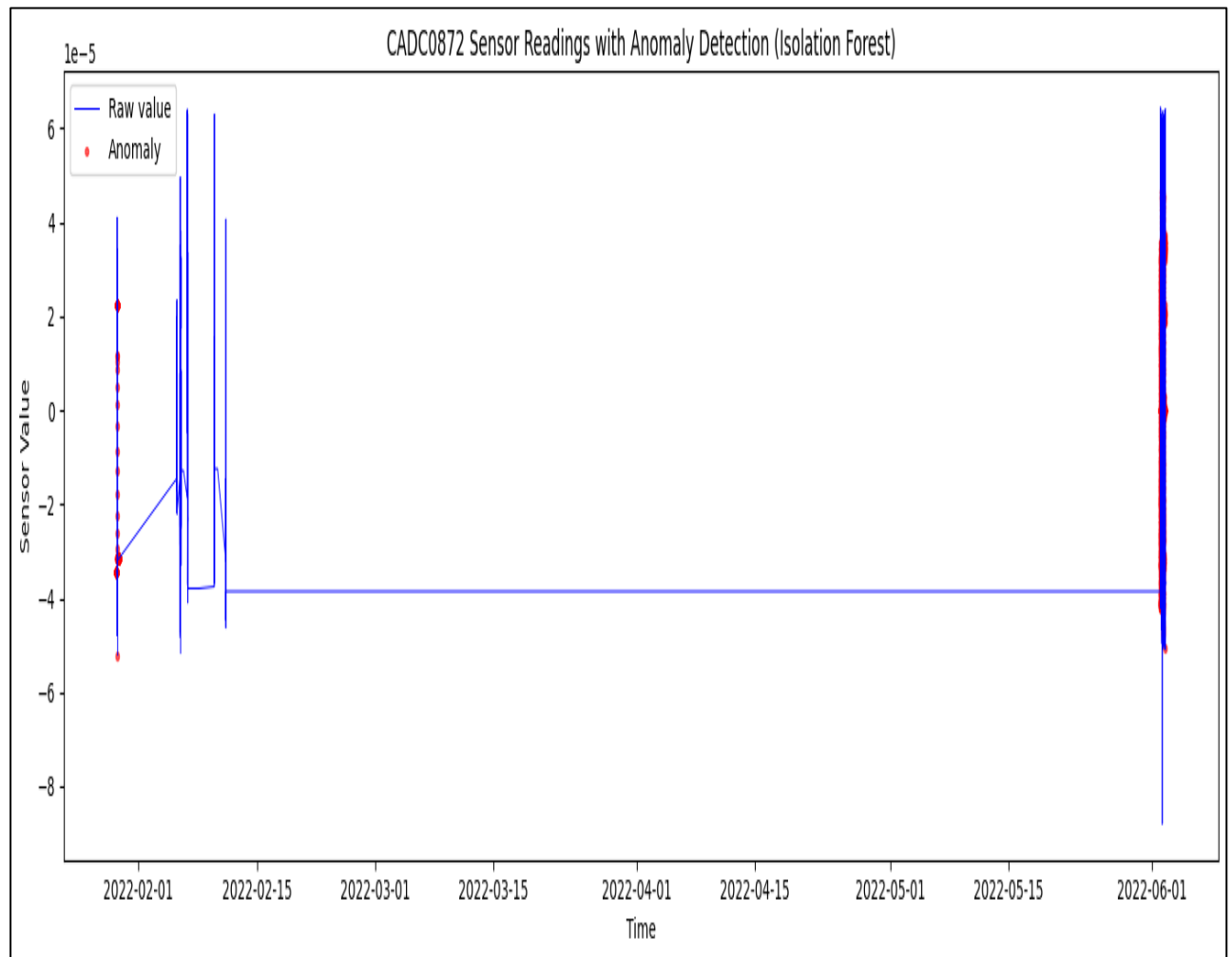
2022-01-29 09:34:57+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.121837
2022-01-29 09:35:02+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.229221
2022-01-29 09:35:07+00:00	-3.4351e-05	-1.63882e-05	-4.48813e-05	0.0	0.0	0.038572	0.0115385	0.0	0.347797

“Table A.1: First 100 rows of the cleaned, pivoted telemetry data.”

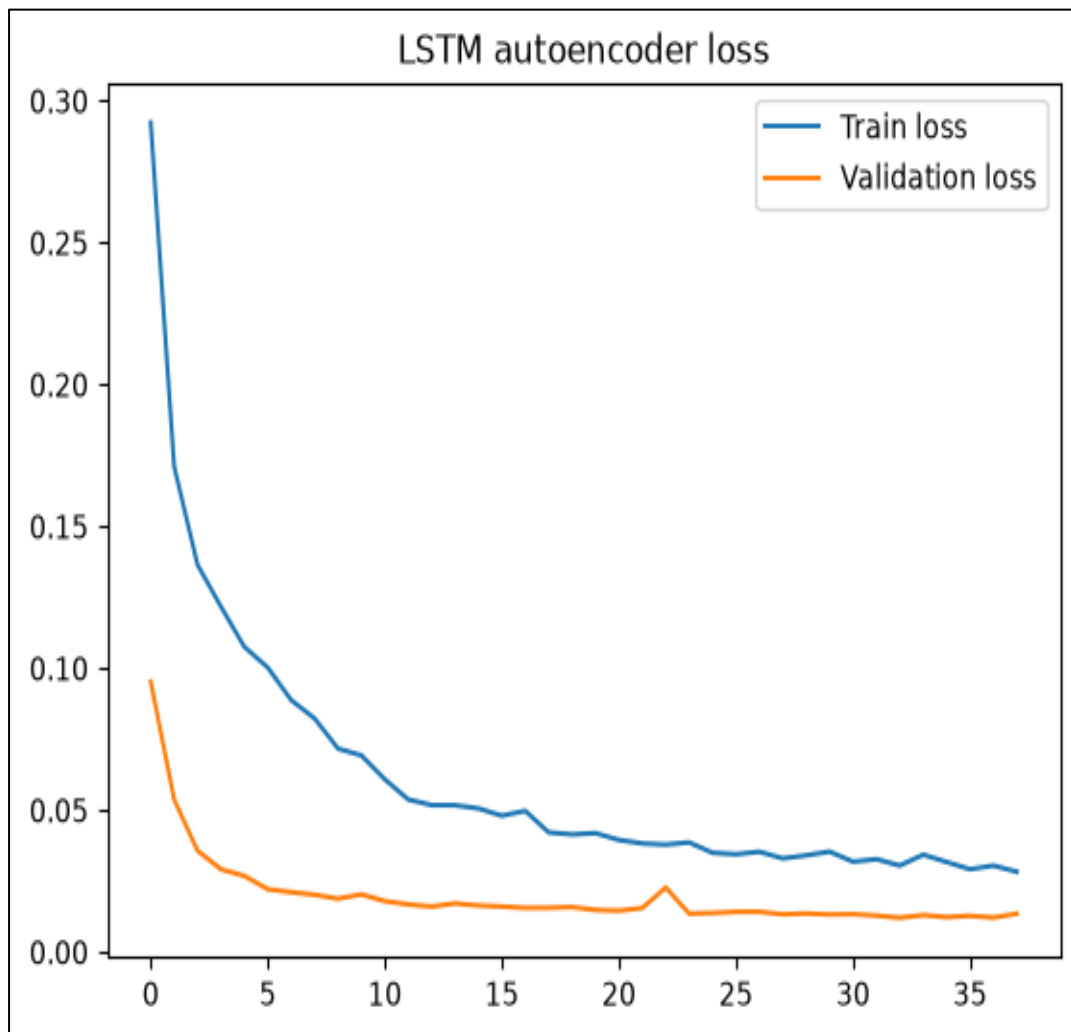
(b) Data Visualization (screenshots – 4 pages)

Including four full-page screenshots:

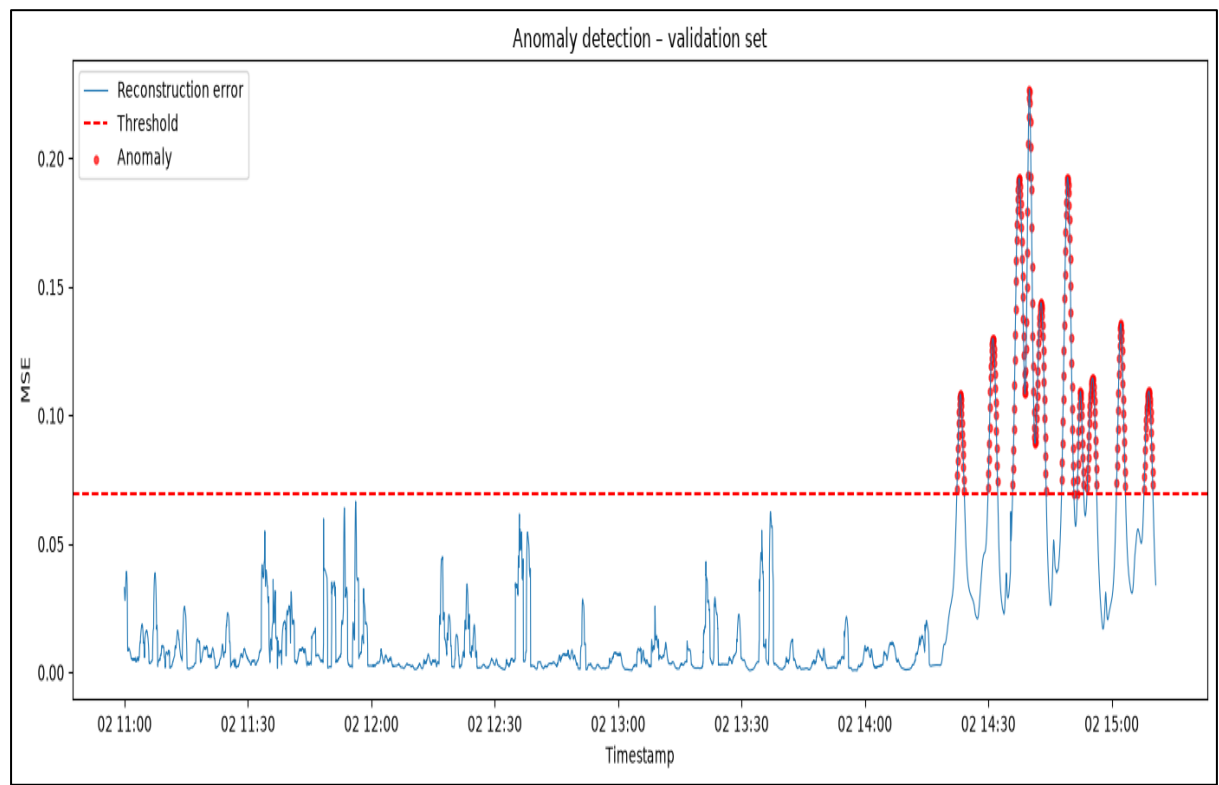
- Page 1: Anomaly overlay plot (Isolation Forest)



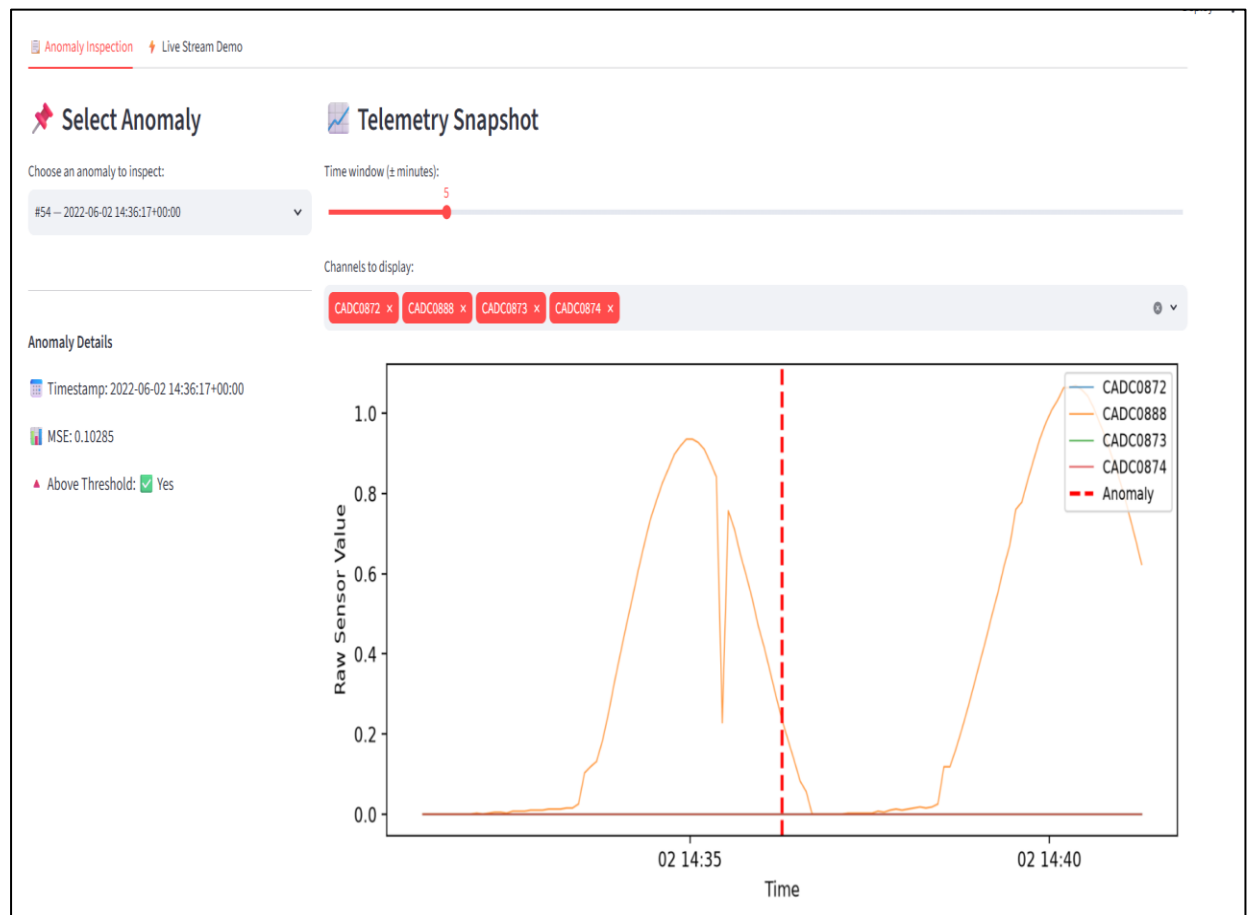
- Page 2: LSTM loss curves



- Page 3: Reconstruction error plot



- Page 4: Dashboard telemetry snapshot



(c) Coding Algorithm Implementation (3 pages)

Providing key code snippets that best represent the project's core logic, for example:

- Page 1: Data preprocessing and feature engineering (pivot, rolling stats)

Source: satguard_eda.ipynb – the cell where I pivot, forward-fill, and compute rolling statistics.

```
# Pivot raw telemetry into multivariate time series
ts = df.pivot_table(index='timestamp', columns='channel',
values='value')
ts.ffill(inplace=True)
ts.dropna(inplace=True)

# Feature engineering - rolling statistics per channel
roll_mean = ts.rolling(window=50,
min_periods=1).mean().add_suffix(' mean')
roll_std = ts.rolling(window=50,
min_periods=1).std().add_suffix(' std')
roll_min = ts.rolling(window=50,
min_periods=1).min().add_suffix(' min')
roll_max = ts.rolling(window=50,
min_periods=1).max().add_suffix(' max')
ts_features = pd.concat([ts, roll_mean, roll_std, roll_min, roll_max],
axis=1)
ts_features.dropna(inplace=True)
print("Feature matrix shape:", ts_features.shape)
```

Code C.1 – Multi-subsystem feature fusion: raw telemetry is pivoted, missing values filled, and rolling statistics (mean, std, min, max) are added to create 45 features per timestamp.

- Page 2: LSTM-Autoencoder model definition and training loop

Source: lstm_autoencoder.ipynb – the cell that builds and compiles the model

(The model architecture definition and training call).

```
# Build LSTM Autoencoder
input_dim = 9    # 9 sensor channels
inputs = Input(shape=(SEQ_LEN, input_dim))
encoder = LSTM(64, activation='tanh', return_sequences=False)(inputs)
bottleneck = Dense(32, activation='relu')(encoder)
decoder = RepeatVector(SEQ_LEN)(bottleneck)
decoder = LSTM(64, activation='tanh', return_sequences=True)(decoder)
outputs = TimeDistributed(Dense(input_dim))(decoder)

autoencoder = Model(inputs, outputs)
autoencoder.compile(optimizer='adam', loss='mse')
autoencoder.summary()

# Train with early stopping
early_stop = EarlyStopping(monitor='val_loss', patience=5,
                           restore_best_weights=True)
history = autoencoder.fit(X_train, X_train, epochs=50, batch_size=128,
                        validation_data=(X_val, X_val),
                        callbacks=[early_stop])
```

Code C.2 – LSTM-Autoencoder architecture: encoder compresses 60-timestep windows into a latent vector; decoder reconstructs the original input. Trained to minimise reconstruction MSE with early stopping.

- Page 3.1: RAG pipeline function (generate_report())

RAG function (from rag_llm_diagnostics.ipynb):

```
def generate_report(anomaly_description, retriever):
    docs = retriever.invoke(anomaly_description)
    context = "\n".join([d.page_content for d in docs])
    prompt = f"""You are a satellite fault diagnostic assistant.
Based on the anomaly description
and the retrieved fault recovery guidelines, write a diagnostic
report with:
    **Anomaly Snapshot**, **Probable Cause**, **Recommended
Actions**, **Confidence Note**.
```

Code C.3.1 – RAG pipeline: the anomaly description is embedded, top-2 relevant chunks are retrieved from ChromaDB, and a locally hosted Llama 3 model generates a structured diagnostic report.

Page 3.2 : Kafka producer/consumer script :

Kafka streaming demo (from kafka_streaming_demo.py):

```
# Producer loop
for i in range(windows):
    window = data_scaled[i:i+SEQ_LEN].tolist()
    timestamp = str(ts.index[i+SEQ_LEN-1])
    producer.send(TOPIC, {"window_index": i, "window": window,
"timestamp": timestamp})

# Consumer loop
for msg in consumer:
    window = np.array(msg.value["window"])
    X_pred = autoencoder.predict(window.reshape(1, SEQ_LEN, 9),
verbose=0)
    mse = np.mean(np.square(window - X_pred))
    if mse > THRESHOLD:
        print(f"🚨 Anomaly at {msg.value['timestamp']} | MSE:
{mse:.5f}")
```

Code C.3.2 – Kafka streaming proof-of-concept: a producer sends telemetry windows to a topic; a consumer runs the same LSTM-Autoencoder to detect anomalies in real time.

(d) Setup and Execution Commands

d.1 Start the Streamlit Dashboard

Open a terminal in the project folder and run:

```
py -3.11 -m streamlit run app.py
```

Then open <http://localhost:8501> in a browser.

d.2 Run a Jupyter Notebook (EDA, LSTM, RAG)

```
cd SatGuard-AI
```

```
py -3.11 -m jupyter notebook
```

d.3 Run the Kafka Streaming Proof-of-Concept

(Ensure Docker Desktop is running)

```
docker compose up -d          # start Kafka + Zookeeper
```

```
py -3.11 kafka_streaming_demo.py    # run producer/consumer
```

```
docker compose down          # stop Kafka
```

d.4 Build and Run the Docker Container

```
docker build -t satguard .      # build the image
```

```
docker run -p 8501:8501 satguard  # start the container
```

Open <http://localhost:8501> in a browser.

(If localhost fails, use the container's internal IP, e.g. <http://172.17.0.2:8501>)

d.5 Compute Evaluation Metrics

Run the evaluation cell inside `lstm_autoencoder.ipynb` (or `evaluation_metrics.ipynb`).

The metrics are saved to `evaluation_metrics.json` and displayed in the dashboard.

d.6 RAG Retrieval Baseline Evaluation

```
py -3.11 rag_evaluation.py
```

Results are saved to `rag_evaluation_results.json`.

References

- **Hundman, K., et al. (2018).** “Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Dynamic Thresholding.” KDD 2018.
- **Gao, Y., et al. (2020).** “Anomaly Detection for Telemetry Time Series Using Deep Learning.” IEEE Access, vol. 8, pp. 61936-61948.
- **Malhotra, P., et al. (2016).** “LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection.” ICML Workshop 2016.
- **Sakurada, M., et al. (2014).** “Anomaly Detection Using Autoencoders with Nonlinear Dimensionality Reduction.” MLSDA Workshop 2014. (optional)
- **Lewis, P., et al. (2020).** “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” NeurIPS 2020.
- **Borgeaud, S., et al. (2022).** “Improving Language Models by Retrieving from Trillions of Tokens.” ICML 2022.